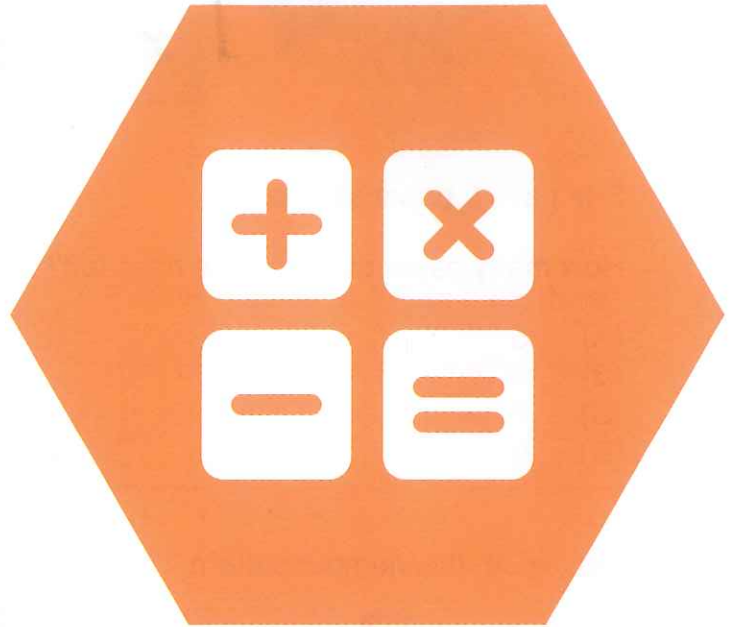




UNSW Global
AUSTRALIA

PAPER A



2015 ICAS

International Competitions
and Assessments for Schools

MATHEMATICS

Educational Assessment Australia
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DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

40 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

There are **40 MULTIPLE-CHOICE QUESTIONS** (1–40).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a ruler and spare paper.

You are **NOT** allowed to use a calculator.

1. Lisa made these paper stars.



She gave 2 paper stars away.

How many paper stars did she have left?

- (A) 5
- (B) 6
- (C) 7
- (D) 9

2. Ajay wrote this number pattern.

30, 25, 20, 15, ?

What number must ? be?

- (A) 14
- (B) 10
- (C) 5
- (D) 0

3. Which object is most like a cylinder?



(A)



(B)

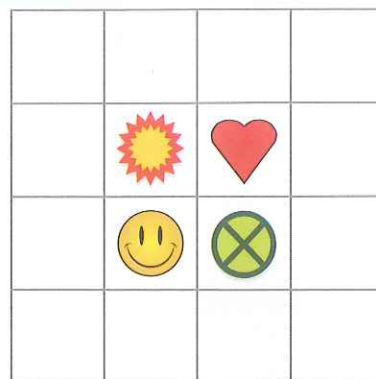


(C)



(D)

4. Mia has some pictures.



Which picture is in the third row from the top and second column from the left?



(A)



(B)



(C)



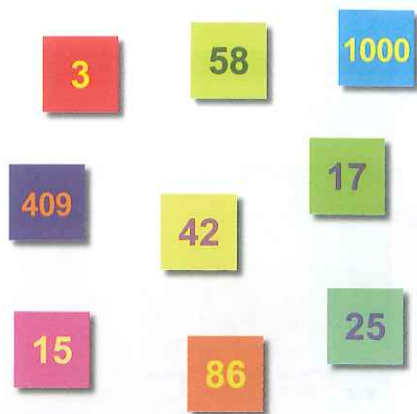
(D)

5. Chandra is going to measure the length of the school playground.

Which unit should Chandra use?

- (A) litres
- (B) metres
- (C) kilograms
- (D) square metres

6. Raj has these number cards.



How many cards have an even number written on them?

- (A) 2
- (B) 3
- (C) 4
- (D) 5

8. Here is a picture of baby Julie.



Belle is 3 times as tall as baby Julie.

How tall is Belle in centimetres (cm)?

- (A) 150
- (B) 100
- (C) 53
- (D) 50

7. On Friday, Molly thought:



What is the chance that Molly is correct?

- (A) impossible
- (B) unlikely
- (C) likely
- (D) certain

9. Mary read the time on the clock.

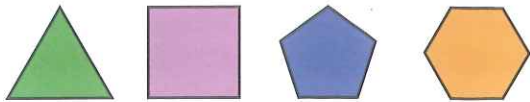


Her class starts in 10 minutes.

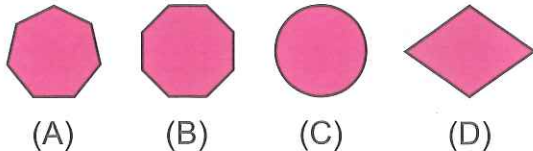
What time does Mary's class start?

- (A) 6:08
- (B) 10:40
- (C) 11:16
- (D) 11:40

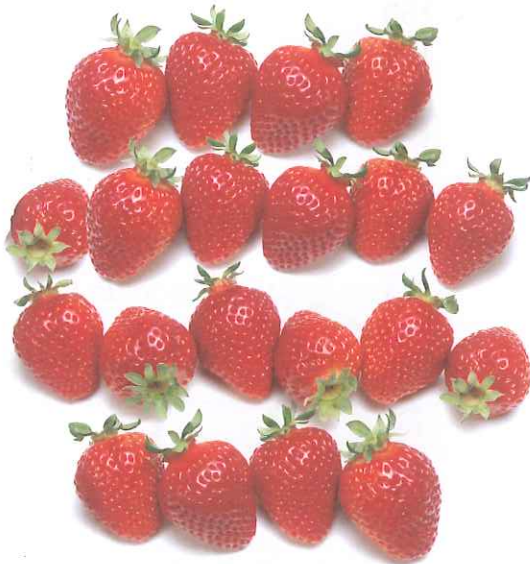
10. Adam is making a pattern using shapes.



What must the next shape be in Adam's pattern?



11. Hans had 20 strawberries.
He gave his sister one-quarter of them.



How many strawberries did Hans give his sister?

- (A) 4
(B) 5
(C) 15
(D) 25

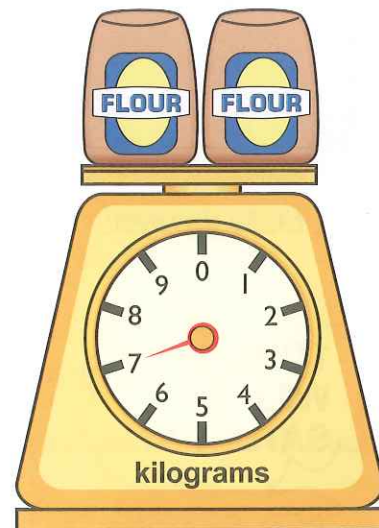
12. Jane and Heidi played basketball.
Each goal was worth 2 points.

Jane scored 3 goals and Heidi scored 5 goals.

How many points did they score altogether?

- (A) 8
(B) 10
(C) 15
(D) 16

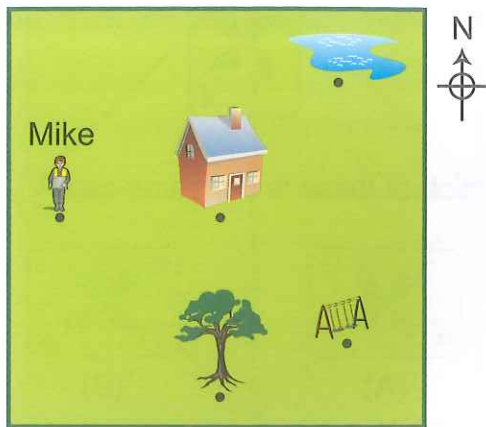
13. This picture shows the mass of 2 bags of flour of flour.



What is the mass of 1 bag of flour, in kilograms?

- (A) $3\frac{1}{2}$ (B) $4\frac{1}{2}$
(C) 7 (D) 14

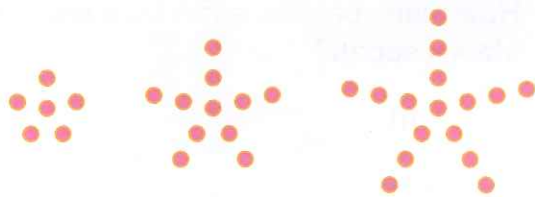
14. From where Mike is standing, he walks in a straight line to the tree.



In which direction does Mike walk?

- (A) north-east
- (B) north-west
- (C) south-east
- (D) south-west

15. Vijay drew a pattern of shapes made up of dots.

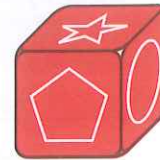


Shape 1 Shape 2 Shape 3

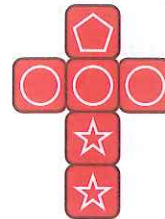
How many dots will be in Shape 4?

- (A) 21
- (B) 24
- (C) 25
- (D) 27

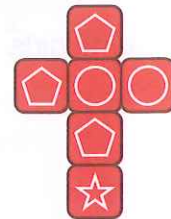
16. Jenny wants to make a dice.



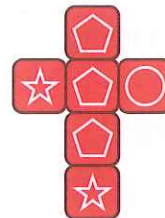
Which shape will fold to make a dice that gives Jenny the best chance of rolling a star?



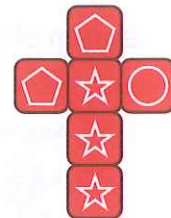
(A)



(B)

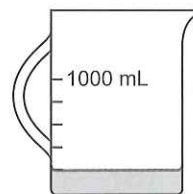


(C)

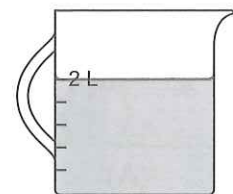


(D)

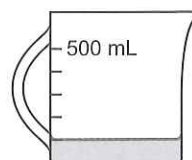
17. Which jug contains exactly 200 mL of liquid?



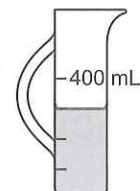
(A)



(B)



(C)



(D)

18. There were 40 students in a dance group. Each student wore only one costume. The costumes came in three colours.

Costume colour	Boys	Girls
black	4	2
gold	8	9
red	5	?

How many girls wore a red costume?

- (A) 7
(B) 11
(C) 12
(D) 28

19. The diagram shows Tim flying a kite.



Which kite is flying at about the same angle as Tim's?



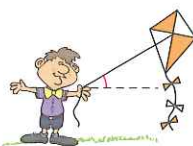
(A)



(B)

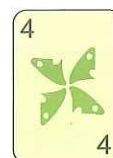


(C)



(D)

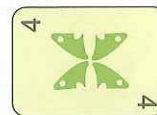
20. This is Jake's card.



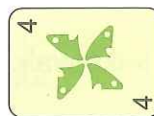
Which of these is the same card?



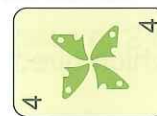
(A)



(B)



(C)



(D)

21. Mark told only James his secret. James told his friends, Pete and Andy. Pete and Andy each told three different people.

How many people, apart from Mark, know Mark's secret?

- (A) 10
(B) 9
(C) 8
(D) 6

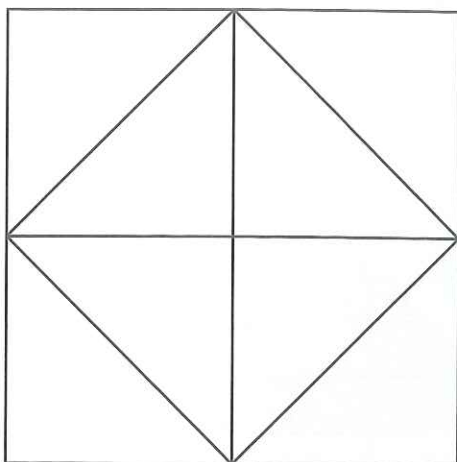
22. Jane is counting up by threes.

She starts to write this pattern in four columns as shown.

(A)	(B)	(C)	(D)
3	6	9	12
15	18	21	24
27			

In which column should Jane write 42?

23. How many triangles are in this picture?



- (A) 12
- (B) 10
- (C) 8
- (D) 4

24. Polly's scoop holds 80 grams of flour. She fills her jar with 5 full scoops of flour.



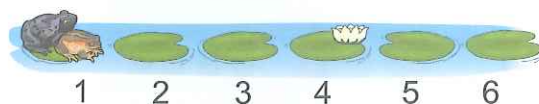
How much more flour does the jar hold than the scoop?

- (A) 64 grams
- (B) 75 grams
- (C) 320 grams
- (D) 400 grams

25. $603 - 75 = ?$

- (A) 528
- (B) 538
- (C) 628
- (D) 672

26. Two frogs sit on a lily pad as shown.



The large frog jumps forward 5 lily pads at a time. The small frog jumps forward 2 lily pads at a time.



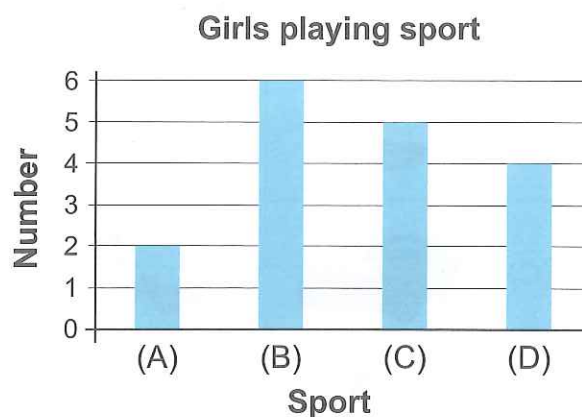
What is the number of the first lily pad that both frogs will land on?

- (A) 7
- (B) 9
- (C) 10
- (D) 11

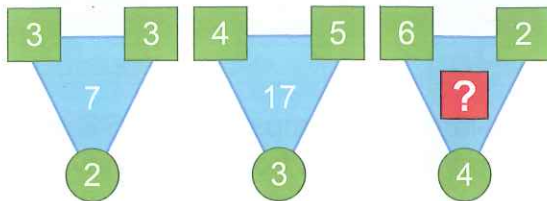
27. Each girl in Sasha's class plays only one sport.

Three more girls play tennis than play soccer.

Which column in this graph shows the girls who play tennis?



28. The numbers in each diagram follow the same pattern.



What number must **?** be?

- (A) 4
(B) 8
(C) 26
(D) 27

29. Lin and her friends estimated the mass of a banana, a car and a baby.



Which option shows the best estimates?

	About 100 grams	About 10 kilograms
(A)		
(B)		
(C)		
(D)		

30. Minh stacked these boxes for a display in his shop.

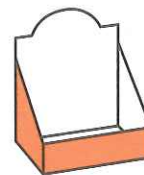
The display looks the same from the back as it does from the front.



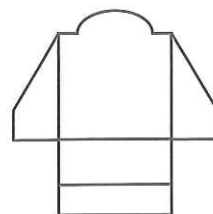
How many boxes have their faces touching only 3 other boxes?

- (A) 2
(B) 4
(C) 6
(D) 14

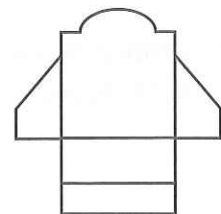
31. This is a display box.



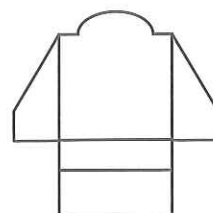
Which of the following is a net of this box?



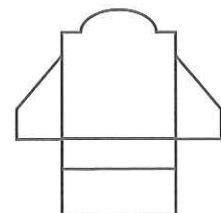
(A)



(B)

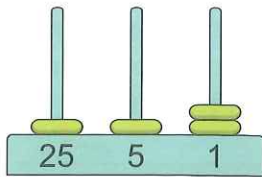


(C)

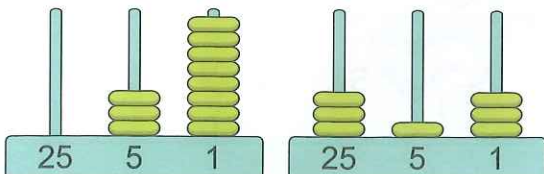


(D)

32. The beads on this abacus make up the number 32.

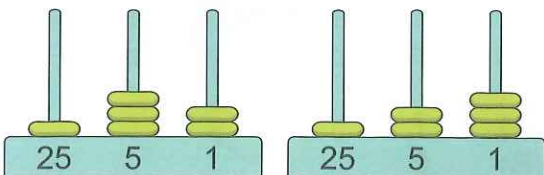


On which abacus do the beads make up the number 38?



(A)

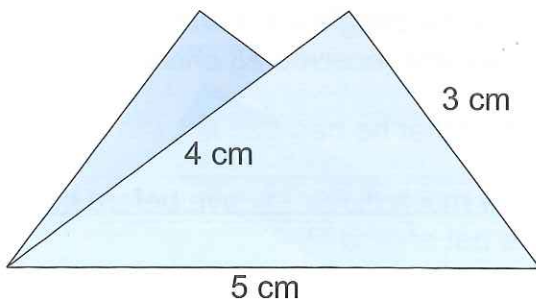
(B)



(C)

(D)

33. Luisa had a rectangular piece of paper. She folded it in half as shown.

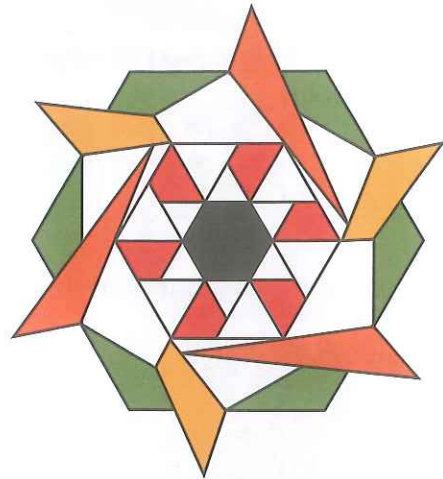


NOT TO SCALE

What is the perimeter of the unfolded piece of paper, in centimetres?

- (A) 12
(B) 14
(C) 18
(D) 19

34. Joe drew a picture.

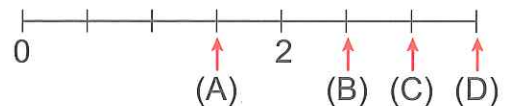


How many lines of symmetry does his picture have?

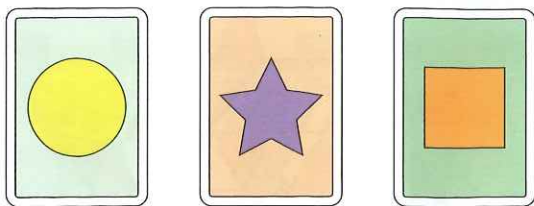
- (A) 0
(B) 2
(C) 3
(D) 6

35. Jack has to complete the labelling on this number line.

Where must he write the number 3?

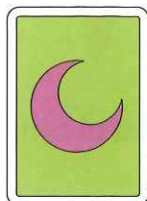


36. Andy had these three cards.



He could pair these cards in three ways.

Andy got one new card.



How many ways could he pair these four cards?

- (A) 8
- (B) 6
- (C) 4
- (D) 3

37. In Anna's school outstanding students win medals. For each medal they win, their class gets some points.

The table shows the number of points for each type of medal.

Medal	Points
 GOLD	3
 SILVER	2

Anna's class got 19 points.

What is the maximum number of medals won by students in Anna's class?

- (A) 7
- (B) 8
- (C) 9
- (D) 19

38. Arun bought a ball with a \$20 note and received \$7 change.

Later, he bought a bat with three \$10 notes and received \$8 change.

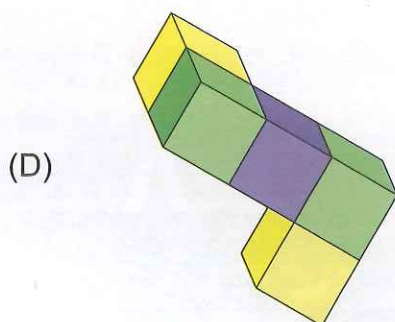
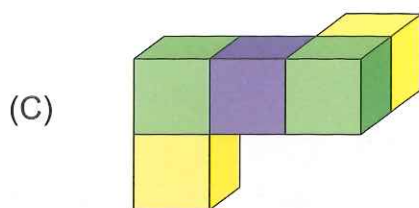
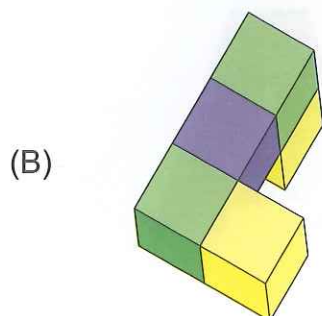
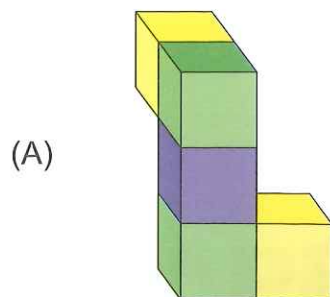
Altogether he had \$25 left.

How much did Arun have before he bought the bat and ball?

- (A) \$76
- (B) \$75
- (C) \$70
- (D) \$60

39. Jamie made a solid using cubes. Three of these options are pictures of Jamie's solid and one is not.

Which one is **NOT**?



40. Dedi had a combination lock. To open the lock he started at 37.

He made three half turns clockwise and three quarter turns anticlockwise.



What number did he end up at?

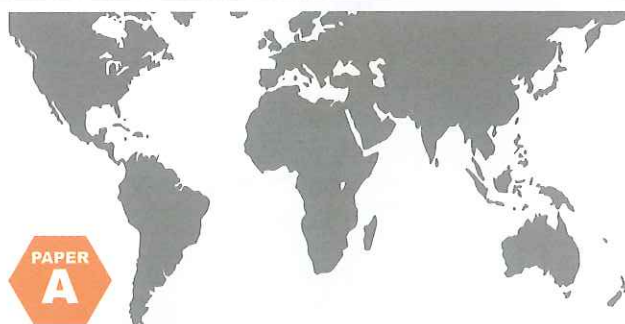
- (A) 7
(B) 17
(C) 27
(D) 37

Acknowledgment

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The following year levels should sit THIS Paper:

Australia	Year 3
Brunei	Primary 3
Indonesia	Year 4
Malaysia	Standard 3
New Zealand	Year 4
Pacific Region	Year 4
Singapore	Primary 2
South Africa	Grade 3



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HOW TO FILL OUT THIS SHEET:



- Print your details clearly in the boxes provided.
- Make sure you fill in only one oval in each column.
- Rub out all mistakes completely.

EXAMPLE 1: Debbie Bach

FIRST NAME

LAST NAME

DEBBIE

BACH

EXAMPLE 2: Chan Ai Beng

FIRST NAME

LAST NAME

C	H	A	N	A	I		B	E	N	G
A	A		A		A	A	A	A	A	A
B	B	B	B	B	B	B		B	B	B
	C	C		C	C	C	C	C	C	C
D	D			D	D	D	D	D	D	D
E				E	E	E	E			

EXAMPLE 3: Jamal bin Abas

FIRST NAME

LAST NAME

J	A	M	A	L		B	I	N	A	B	A	S
☐	☐	☐	☐	☐	☒	☐	☐	☐	☐	☐	☐	☐
A	☒	A	☒	A	A	A	A	A	☒	A	☒	A
B	B	B	B	B	B	B	☒	B	B	☒	B	B
C	C	C	C	C	C	C	C	C	C	C	C	C
D	D	D	D	D	D	D	D	D	D	D	D	D
E	E	E	E	E	E	E	E	E	E	E	E	E

FIRST NAME to appear on certificate

[illegible]

LAST NAME to appear on certificate

[illegible]

Are you male or female?

☐ Male ☐ Female



Does anyone in your home usually speak a language other than English?

☐ Yes ☐ No

School name: _____

Town / suburb: _____

Today's date: / /

Postcode: _____

DATE OF BIRTH

Day		Month		Year	
0	0	0	0	0	0
1	1	1	1	1	1
2	2	2	2	2	2
3	3	3	3	3	3
4	4	4	4	4	4
5	5	5	5	5	5
6	6	6	6	6	6
7	7	7	7	7	7
8	8	8	8	8	8
9	9	9	9	9	9

CLASS
(optional)

	(A)	(K)
	(B)	(L)
	(C)	(M)
	(D)	(N)
	(E)	(O)
	(F)	(P)
	(G)	(Q)
	(H)	(R)
	(I)	(S)
	(J)	(T)

TO ANSWER THE QUESTIONS

MULTIPLE CHOICE

Questions 1 to 40

Example: $4 + 6 =$

- (A) 2
- (B) 9
- (C) 10
- (D) 24

The answer is 10, so fill in the oval **C**, as shown.

A **B** **C** **D**



START

- 1 **A** **B** **C** **D**
- 2 **A** **B** **C** **D**
- 3 **A** **B** **C** **D**
- 4 **A** **B** **C** **D**
- 5 **A** **B** **C** **D**
- 6 **A** **B** **C** **D**
- 7 **A** **B** **C** **D**
- 8 **A** **B** **C** **D**
- 9 **A** **B** **C** **D**
- 10 **A** **B** **C** **D**

- 11 **A** **B** **C** **D**
- 12 **A** **B** **C** **D**
- 13 **A** **B** **C** **D**
- 14 **A** **B** **C** **D**
- 15 **A** **B** **C** **D**
- 16 **A** **B** **C** **D**
- 17 **A** **B** **C** **D**
- 18 **A** **B** **C** **D**
- 19 **A** **B** **C** **D**
- 20 **A** **B** **C** **D**

- 21 **A** **B** **C** **D**
- 22 **A** **B** **C** **D**
- 23 **A** **B** **C** **D**
- 24 **A** **B** **C** **D**
- 25 **A** **B** **C** **D**
- 26 **A** **B** **C** **D**
- 27 **A** **B** **C** **D**
- 28 **A** **B** **C** **D**
- 29 **A** **B** **C** **D**
- 30 **A** **B** **C** **D**

- 31 **A** **B** **C** **D**
- 32 **A** **B** **C** **D**
- 33 **A** **B** **C** **D**
- 34 **A** **B** **C** **D**
- 35 **A** **B** **C** **D**
- 36 **A** **B** **C** **D**
- 37 **A** **B** **C** **D**
- 38 **A** **B** **C** **D**
- 39 **A** **B** **C** **D**
- 40 **A** **B** **C** **D**

MATHS A 2015 – ANSWER SHEET

Question Number	Paper A
1	A
2	B
3	B
4	C
5	B
6	C
7	D
8	A
9	D
10	A
11	B
12	D
13	A
14	C
15	A
16	D
17	A
18	C
19	A
20	D
21	B
22	B
23	A
24	C
25	A
26	D
27	C
28	B
29	B
30	C

Question Number	Paper A
31	B
32	D
33	B
34	A
35	C
36	B
37	C
38	D
39	D
40	A